

Phetolo Tshukudu

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PROFESSIONAL SUMMARY

Final-year BSc Mathematical Sciences student at the University of Johannesburg, majoring in Computer Science and Mathematical Statistics, with a strong foundation in software engineering, backend development and data analysis. Hands-on experience building Java/Spring Boot REST APIs and Python data-analysis applications, with practical experience in relational databases, object-oriented programming, automated testing and statistical analysis. Interested in graduate and junior software engineering, backend development and data engineering opportunities where I can apply strong programming, analytical and problem-solving skills to build reliable, data-driven systems.

EDUCATION

University of Johannesburg

2023 - Present

BSc. Mathematical Sciences in Computer Science and Mathematical Statistics

Relevant coursework

Object-Oriented Programming • Data Structures & Algorithms • Database Systems • Software Engineering • Computer Networks • Operating Systems • Mathematical Statistics • Statistical Modelling • Regression Analysis

Schooling /Career Highlights

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- High school learner council
 - Academic award in Mathematics
 - UJ UJenius Certificate

SKILLS

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- **Languages:** Java, Python ,SQL, R
 - **Backend:** Spring Boot, Spring Security, REST APIs, JWT, JPA/Hibernate
 - **Python & Data:** Pandas, Data Cleaning, Data Analysis, Statistical Analysis, Dataclasses
 - **Databases:** MySQL, SQLite, Relational Database Design
 - **Testing:** Pytest, JUnit 5, Mockito
 - **Development Tools:** Git, GitHub, Maven, Eclipse IDE, Postman, MySQL Workbench
 - **Core Computer Science:** Data Structures & Algorithms, Graph Algorithms, Database Systems, Software Engineering
 - **Statistics & Data:** Statistical Modelling, Regression Analysis, Data Analysis, Data Interpretation & Visualisation, Model Evaluation & Validation

WORK EXPERIENCE (PART TIME WORK)

MATHEMATICS & PHYSICAL SCIENCES TUTOR Ghetto Hero Saturday School | Jun 2023 – Jul 2023

- Tutored Grade 10–12 learners in Mathematics and Physical Sciences.

SELECTED PROJECTS

Phetolo Tshukudu | Backend development and data analysis

COMMUNITY INFRASTRUCTURE REPORTING API

Java 21 | Spring Boot | Spring Security | JWT | BCrypt | MySQL | Flyway | Docker

- Built a secure API where residents report infrastructure issues and workers/admins review incidents through verification, work and resolution.
- Implemented role-based access, coordinate validation, filtering and analytics; containerised the API and MySQL with Docker Compose, health checks and persistent storage.
- Recorded the author and timestamp of status changes; used optimistic locking to reject conflicting updates and roll back associated history.

github.com/phetolo-tshukudu/infrastructure-reporting

FINCORE - PERSONAL FINANCE BACKEND

Java 21 | Spring Boot | JWT | PostgreSQL | Flyway | Docker | OpenAPI

- Built a REST API for managing users, accounts, transactions and personal budgets.
- Implemented authentication, persistence, schema migrations, validation, API documentation and Docker support.

github.com/phetolo-tshukudu/fincore

FINANCIAL DATA ANALYZER

Python | Pandas | Pytest | Dataclasses | Statistics

- Built an application for cleaning, processing and analysing financial transaction data.
- Added spending statistics, budget tracking and 16 automated pytest tests for core business logic.

PERSONALFINANCE API - FINANCIAL MANAGEMENT BACKEND

Java | Spring Boot | Spring Security | JWT | MySQL | JPA/Hibernate | JUnit 5 | Mockito

- Developed a layered REST API for users, accounts, transactions and budgets.
- Implemented security, persistence, validation, exception handling and unit tests.

github.com/phetolo-tshukudu/finance-api

WILDFIRE SIMULATOR

Java | JavaFX | Graph Algorithms | Dijkstra's Algorithm

- Collaborated on an application that simulates wildfire spread and identifies safer escape routes.
- Used Dijkstra's algorithm, priority queues and custom data structures to model routes and fire conditions.

github.com/phetolo-tshukudu/Wildfire-Simulator